

SOLIDWORKS API (.NET VBA C#) — Reference

(277/337) ISketchRelationManager

SOLIDWORKS API Documentation (converted from CHM)

ISketchRelation (Type)

ISketchRelation Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISketchRelation Interface

Provides access to the entities for a sketch relation.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ISketchRelation
```

Visual Basic (Usage)

```
Dim instance As ISketchRelation
```

C#

```
public interface ISketchRelation
```

C++/CLI

```
public interface class ISketchRelation
```

Visual Basic for Applications (VBA) Syntax

See [SketchRelation](#).

Example

[Replace Sketch Relation \(VBA\)](#)
[Get Sketch Relations \(VBA\)](#)
[Get Sketch Relations \(VB.NET\)](#)
[Get Sketch Relations \(C#\)](#)

Accessors

[ISketchPoint::GetRelations](#) and [ISketchPoint::!GetRelations](#)
[ISketchRelationManager::AddRelation](#) and [ISketchRelation::!AddRelation](#)
[ISketchRelationManager::GetRelations](#) and [ISketchRelationManager::!GetRelations](#)
[ISketchSegment::GetRelations](#) and [ISketchSegment::!GetRelations](#)

Access Diagram

[SketchRelation](#)

See Also

[ISketchRelation Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISketchRelationManager Interface](#)

ISketchRelation Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISketchRelation Interface

The following tables list the members exposed by [ISketchRelation](#).

Public Properties

Name	Description
 Suppressed	Gets or sets whether this relation is suppressed or not.

[Top](#)

Public Methods

Name	Description
 GetDefinitionEntities	Obsolete. Superseded by ISketchRelation::GetDefinitionEntities2
 GetDefinitionEntities2	Gets the actual entities associated with this sketch relation.
 GetDisplayDimension	Gets the display dimension object for this sketch relation.
 GetEntities	Gets the entities associated with this sketch relation.
 GetEntitiesCount	Gets the number of entities associated with this sketch relation.
 GetEntitiesType	Gets the types of entities in this sketch relation.
 GetRelationType	Gets the type of sketch relation.
 IGetDefinitionEntities	Obsolete. Superseded by ISketchRelation::IGetDefinitionEntities2
 IGetDefinitionEntities2	Gets the actual entities associated with this sketch relation.
 IGetEntities	Gets the entities associated with this sketch relation.
 IGetEntitiesType	Gets the types of entities in this sketch relation.
 ReplaceEntity	Replaces an entity in this sketch relation.

[Top](#)

See Also

[ISketchRelation Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISketchRelationManager Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISketchRelation Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISketchRelation Interface

For a list of all members of this type, see [ISketchRelation members](#).

Public Methods

Name	Description
 GetDefinitionEntities	Obsolete. Superseded by ISketchRelation::GetDefinitionEntities2
 GetDefinitionEntities2	Gets the actual entities associated with this sketch relation.
 GetDisplayDimension	Gets the display dimension object for this sketch relation.
 GetEntities	Gets the entities associated with this sketch relation.
 GetEntitiesCount	Gets the number of entities associated with this sketch relation.
 GetEntitiesType	Gets the types of entities in this sketch relation.
 GetRelationType	Gets the type of sketch relation.
 IGetDefinitionEntities	Obsolete. Superseded by ISketchRelation::IGetDefinitionEntities2
 IGetDefinitionEntities2	Gets the actual entities associated with this sketch relation.
 IGetEntities	Gets the entities associated with this sketch relation.
 IGetEntitiesType	Gets the types of entities in this sketch relation.
 ReplaceEntity	Replaces an entity in this sketch relation.

[Top](#)

See Also

[ISketchRelation Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISketchRelationManager Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISketchRelation Interface Properties

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : ISketchRelation Interface

For a list of all members of this type, see [ISketchRelation members](#).

Public Properties

Name	Description
 Suppressed	Gets or sets whether this relation is suppressed or not.

[Top](#)

See Also

[ISketchRelation Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISketchRelationManager Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetDefinitionEntities Method (ISketchRelation)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRelation Interface](#) : GetDefinitionEntities Method (ISketchRelation)

Obsolete. Superseded by [ISketchRelation::GetDefinitionEntities2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetDefinitionEntities() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchRelation  
Dim value As System.Object  
  
value = instance.GetDefinitionEntities()
```

C#

```
System.object GetDefinitionEntities()
```

C++/CLI

```
System.Object^ GetDefinitionEntities();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of the actual entities for this sketch relation

Visual Basic for Applications (VBA) Syntax

See [SketchRelation::GetDefinitionEntities](#).

Remarks

When you created a sketch relation, an internal entity may have also been created to define that sketch relation. For example, if you added a sketch relation by specifying or selecting a point and an edge, a sketch segment may have been created. If this happened, then

[ISketchRelation::GetEntities](#) and [ISketchRelation::IGetEntities](#) would return the point and sketch segment. To get the actual entities used to define the sketch relation, call [ISketchRelation::GetDefinitionEntities](#) or [ISketchRelation::IGetDefinitionEntities](#).

Because some of the objects may be NULL, you should check for this before accessing them. You should deallocate any dynamically allocated memory.

See Also

[ISketchRelation Interface](#)
[ISketchRelation Members](#)
[ISketchRelation::GetEntitiesCount Method](#)

Availability

SOLIDWORKS 2004 SP4, Revision Number 12.4

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetDefinitionEntities2 Method (ISketchRelation)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRelation Interface](#) : GetDefinitionEntities2 Method (ISketchRelation)

Gets the actual entities associated with this sketch relation.

.NET Syntax

Visual Basic (Declaration)

```
Function GetDefinitionEntities2() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchRelation  
Dim value As System.Object  
  
value = instance.GetDefinitionEntities2()
```

C#

```
System.object GetDefinitionEntities2()
```

C++/CLI

```
System.Object^ GetDefinitionEntities2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of the actual entities for this sketch relation

Visual Basic for Applications (VBA) Syntax

See [SketchRelation::GetDefinitionEntities2](#).

Example

[Get Sketch Relations \(VBA\)](#)
[Get Sketch Relations \(VB.NET\)](#)
[Get Sketch Relations \(C#\)](#)

Remarks

When you create a sketch relation, an internal entity may be created to define that sketch relation. For example, if you add a sketch relation by specifying or selecting a point and an edge, a sketch segment may be created. If this happens, then [ISketchRelation::GetEntities](#) and [ISketchRelation::IGetEntities](#) return the point and sketch segment.

To get the actual entities used to define the sketch relation, call [ISketchRelation::GetDefinitionEntities2](#) or [ISketchRelation::IGetDefinitionEntities2](#). For example, if the relation is of type [swConstraintType_e.swConstraintType_OFFSETEDGE](#), and the relation is created by selecting a face, then this method returns that face.

Because some of the objects may be null, you should check for null before accessing them. You should deallocate any dynamically allocated memory.

See Also

[ISketchRelation Interface](#)
[ISketchRelation Members](#)
[ISketchRelation::GetEntitiesCount Method \(\)](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetDisplayDimension Method (ISketchRelation)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRelation Interface](#) : GetDisplayDimension Method (ISketchRelation)

Gets the display dimension object for this sketch relation.

.NET Syntax

Visual Basic (Declaration)

```
Function GetDisplayDimension() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchRelation  
Dim value As System.Object  
  
value = instance.GetDisplayDimension()
```

C#

```
System.object GetDisplayDimension()
```

C++/CLI

```
System.Object^ GetDisplayDimension();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IDisplayDimension](#)

Visual Basic for Applications (VBA) Syntax

See [SketchRelation::GetDisplayDimension](#).

Example

[Get Sketch Relations \(VBA\)](#)
[Get Sketch Relations \(VB.NET\)](#)
[Get Sketch Relations \(C#\)](#)

See Also

[ISketchRelation Interface](#)
[ISketchRelation Members](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetEntities Method (ISketchRelation)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRelation Interface](#) : GetEntities Method (ISketchRelation)

Gets the entities associated with this sketch relation.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEntities() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchRelation  
Dim value As System.Object  
  
value = instance.GetEntities()
```

C#

```
System.object GetEntities()
```

C++/CLI

```
System.Object^ GetEntities();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of entities for this sketch relation

Visual Basic for Applications (VBA) Syntax

See [SketchRelation::GetEntities](#).

Example

[Replace Sketch Relation \(VBA\)](#)
[Get Sketch Relations \(VBA\)](#)
[Get Sketch Relations \(VB.NET\)](#)
[Get Sketch Relations \(C#\)](#)

Remarks

When you create a sketch relation, an internal entity may also be created to define that sketch relation. For example, if you add a sketch relation by specifying or selecting a point and an edge, a sketch segment may be created. If this happens, then [ISketchRelation::GetEntities](#) and [ISketchRelation::GetDefinitionEntities](#) return the point and sketch segment. To get the actual entities used to define the sketch relation, call [ISketchRelation::GetDefinitionEntities](#) or [ISketchRelation::IGetDefinitionEntities](#).

Because some of the objects may be NULL, you should check for this before accessing them. You should deallocate any dynamically allocated memory.

See Also

[ISketchRelation Interface](#)
[ISketchRelation Members](#)
[ISketchRelation::GetEntitiesCount Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetEntitiesCount Method (ISketchRelation)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRelation Interface](#) : GetEntitiesCount Method (ISketchRelation)

Gets the number of entities associated with this sketch relation.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEntitiesCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketchRelation
Dim value As System.Integer

value = instance.GetEntitiesCount()
```

C#

```
System.int GetEntitiesCount()
```

C++/CLI

```
System.int GetEntitiesCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of entities for this sketch relation

Visual Basic for Applications (VBA) Syntax

See [SketchRelation::GetEntitiesCount](#).

Example

[Replace Sketch Relation \(VBA\)](#)

Remarks

Call this method before calling [ISketchRelation::IGetDefinitionEntities](#), [ISketchRelation::IGetEntities](#), or [ISketchRelation::IGetEntitiesType](#) to determine the size of array for the entities or types of entities associated with the sketch relation.

See Also

[ISketchRelation Interface](#)
[ISketchRelation Members](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetEntitiesType Method (ISketchRelation)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRelation Interface](#) : GetEntitiesType Method (ISketchRelation)

Gets the types of entities in this sketch relation.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEntitiesType() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchRelation
Dim value As System.Object

value = instance.GetEntitiesType()
```

C#

```
System.object GetEntitiesType()
```

C++/CLI

```
System.Object^ GetEntitiesType();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of types of entities in this sketch relation as defined by [swSketchRelationEntityType_e](#)

Visual Basic for Applications (VBA) Syntax

See [SketchRelation::GetEntitiesType](#).

Example

[Get Sketch Relations \(VBA\)](#)

See Also

[ISketchRelation Interface](#)
[ISketchRelation Members](#)
[ISketchRelation::IGetEntities Method](#)
[ISketchRelation::GetEntitiesCount Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetRelationType Method (ISketchRelation)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRelation Interface](#) : GetRelationType Method (ISketchRelation)

Gets the type of sketch relation.

.NET Syntax

Visual Basic (Declaration)

```
Function GetRelationType() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketchRelation
Dim value As System.Integer

value = instance.GetRelationType()
```

C#

```
System.int GetRelationType()
```

C++/CLI

```
System.int GetRelationType();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Type of sketch relation as defined in [swConstraintType_e](#)

Visual Basic for Applications (VBA) Syntax

See [SketchRelation::GetRelationType](#).

Example

[Get All Sketch Relations and Their Types \(VBA\)](#)
[Replace Sketch Relation \(VBA\)](#)
[Get Sketch Relations \(VBA\)](#)
[Get Sketch Relations \(VB.NET\)](#)
[Get Sketch Relations \(C#\)](#)

See Also

[ISketchRelation Interface](#)
[ISketchRelation Members](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetDefinitionEntities Method (ISketchRelation)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRelation Interface](#) : IGetDefinitionEntities Method (ISketchRelation)

Obsolete. Superseded by [ISketchRelation::IGetDefinitionEntities2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetDefinitionEntities( _  
    ByVal NumEntities As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchRelation  
Dim NumEntities As System.Integer  
Dim value As System.Object
```

```
value = instance.IGetDefinitionEntities(NumEntities)
```

C#

```
System.object IGetDefinitionEntities(  
    System.int NumEntities  
)
```

C++/CLI

```
System.Object^ IGetDefinitionEntities(  
    & System.int NumEntities  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumEntities

Number of actual entities

Return Value

- in-process, unmanaged C++: Pointer to an array of the actual entities for this sketch relation
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

When you created a sketch relation, an internal entity may have also been created to define that sketch relation. For example, if you added a sketch relation by specifying or selecting a point and an edge, a sketch segment may have been created. If this happened, then [ISketchRelation::GetEntities](#) and [ISketchRelation::!GetEntities](#) would return the point and sketch segment. To get the actual entities used to define the sketch relation, call [ISketchRelation::GetDefinitionEntities](#) or [ISketchRelation::!GetDefinitionEntities](#).

Because some of the objects may be NULL, you should check for this before accessing them. You should deallocate any dynamically allocated memory.

Before calling this method, call [ISketchRelation::GetEntitiesCount](#) to get the value for NumEntities.

See Also

[ISketchRelation Interface](#)
[ISketchRelation Members](#)

Availability

SOLIDWORKS 2004 SP4, Revision Number 12.4

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetDefinitionEntities2 Method (ISketchRelation)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRelation Interface](#) : IGetDefinitionEntities2 Method (ISketchRelation)

Gets the actual entities associated with this sketch relation.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetDefinitionEntities2( _  
    ByVal NumEntities As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchRelation  
Dim NumEntities As System.Integer  
Dim value As System.Object  
  
value = instance.IGetDefinitionEntities2(NumEntities)
```

C#

```
System.object IGetDefinitionEntities2(  
    System.int NumEntities  
)
```

C++/CLI

```
System.Object^ IGetDefinitionEntities2(  
& System.int NumEntities  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumEntities

Number of actual entities

Return Value

- In-process, unmanaged C++: Pointer to an array of the actual entities for this sketch relation
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

When you create a sketch relation, an internal entity may also be created to define that sketch relation. For example, if you add a sketch relation by specifying or selecting a point and an edge, a sketch segment may be created. If this happens, then [ISketchRelation::GetEntities](#) and [ISketchRelation::!GetEntities](#) return the point and sketch segment.

To get the actual entities used to define the sketch relation, call [ISketchRelation::GetDefinitionEntities2](#) or [ISketchRelation::!GetDefinitionEntities2](#). For example, if the relation is of type [swConstraintType_e.swConstraintType_OFFSETEDGE](#), and the relation is created by selecting a face, then this method returns that face.

Because some of the objects may be null, you should check for null before accessing them. You should deallocate any dynamically allocated memory.

Before calling this method, call [ISketchRelation::GetEntitiesCount](#) to get the value for NumEntities.

See Also

[ISketchRelation Interface](#)
[ISketchRelation Members](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetEntities Method (ISketchRelation)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRelation Interface](#) : IGetEntities Method (ISketchRelation)

Gets the entities associated with this sketch relation.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetEntities( _  
    ByVal NumEntities As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchRelation  
Dim NumEntities As System.Integer  
Dim value As System.Object  
  
value = instance.IGetEntities(NumEntities)
```

C#

```
System.object IGetEntities(  
    System.int NumEntities  
)
```

C++/CLI

```
System.Object^ IGetEntities(  
& System.int NumEntities  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumEntities
Number of entities

Return Value

- in-process, unmanaged C++: Pointer to an array of entities for this sketch relation
- VBA, VB.NET, C#, and C++/CLI: Not supported
See [In-process Methods](#) for details about this type of method.

Remarks

When you create a sketch relation, an internal entity may also be created to define that sketch relation. For example, if you add a sketch relation by specifying or selecting a point and an edge, a sketch segment may be created. If this happens, then [ISketchRelation::GetEntities](#) and [ISketchRelation::IGetEntities](#) return the point and sketch segment. To get the actual entities used to define the sketch relation, call [ISketchRelation::GetDefinitionEntities](#) or [ISketchRelation::IGetDefinitionEntities](#).

Because some of the objects may be NULL, you should check for this before accessing them. You should deallocate any dynamically allocated memory.

Before calling this method, call [ISketchRelation::GetEntitiesCount](#) to get the value for NumEntities.

See Also

[ISketchRelation Interface](#)
[ISketchRelation Members](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetEntitiesType Method (ISketchRelation)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRelation Interface](#) : IGetEntitiesType Method (ISketchRelation)

Gets the types of entities in this sketch relation.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetEntitiesType( _  
    ByVal NumEntities As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketchRelation  
Dim NumEntities As System.Integer  
Dim value As System.Integer  
  
value = instance.IGetEntitiesType(NumEntities)
```

C#

```
System.int IGetEntitiesType(  
    System.int NumEntities  
)
```

C++/CLI

```
System.int IGetEntitiesType(  
& System.int NumEntities  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumEntities
Number of entities

Return Value

- in-process, unmanaged C++: Pointer to an array of types of entities in this sketch relation as defined by [swSketchRelationEntityTypes_e](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [ISketchRelation::GetEntitiesCount](#) to get the value for NumEntities.

See Also

[ISketchRelation Interface](#)
[ISketchRelation Members](#)
[ISketchRelation::GetEntities Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ReplaceEntity Method (ISketchRelation)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRelation Interface](#) : ReplaceEntity Method (ISketchRelation)

Replaces an entity in this sketch relation.

.NET Syntax

Visual Basic (Declaration)

```
Function ReplaceEntity( _  
    ByVal OldEntity As System.Object, _  
    ByVal NewEntity As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchRelation  
Dim OldEntity As System.Object  
Dim NewEntity As System.Object  
Dim value As System.Boolean
```

```
value = instance.ReplaceEntity(OldEntity, NewEntity)
```

C#

```
System.bool ReplaceEntity(  
    System.object OldEntity,  
    System.object NewEntity  
)
```

C++/CLI

```
System.bool ReplaceEntity(  
& System.Object^ OldEntity,  
& System.Object^ NewEntity  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OldEntity
Entity to replace
NewEntity
Replacement entity

Return Value

True if OldEntity is replaced by NewEntity, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchRelation::ReplaceEntity](#).

Example

[Replace Sketch Relation \(VBA\)](#)

Remarks

This method requires that the sketch be in edit mode.

See Also

[ISketchRelation Interface](#)
[ISketchRelation Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Suppressed Property (ISketchRelation)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRelation Interface](#) : Suppressed Property (ISketchRelation)

Gets or sets whether this relation is suppressed or not.

.NET Syntax

Visual Basic (Declaration)

Property Suppressed As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISketchRelation  
Dim value As System.Boolean
```

```
instance.Suppressed = value
```

```
value = instance.Suppressed
```

C#

```
System.bool Suppressed {get; set;}
```

C++/CLI

```
property System.bool Suppressed {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True the relation is suppressed, false it is not

Visual Basic for Applications (VBA) Syntax

See [SketchRelation::Suppressed](#).

Example

[Replace Sketch Relation \(VBA\)](#)

See Also

[ISketchRelation Interface](#)
[ISketchRelation Members](#)

ISketchRelationManager (Type)

ISketchRelationManager Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISketchRelationManager Interface

Provides access to all sketch relations.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ISketchRelationManager
```

Visual Basic (Usage)



```
Dim instance As ISketchRelationManager
```

C#

```
public interface ISketchRelationManager
```

C++/CLI

```
public interface class ISketchRelationManager
```

Visual Basic for Applications (VBA) Syntax

See [SketchRelationManager](#).

Example

[Replace Sketch Relation \(VBA\)](#)
[Get Sketch Relations \(VBA\)](#)
[Get Sketch Relations \(VB.NET\)](#)
[Get Sketch Relations \(C#\)](#)

Remarks

You can get the associated entities for each sketch relation and vice versa.

Accessors

[ISketch::RelationManager](#)

Access Diagram

[SketchRelationManager](#)

See Also

[ISketchRelationManager Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISketchRelation Interface](#)

ISketchRelationManager Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISketchRelationManager Interface

For a list of all members of this type, see [ISketchRelationManager members](#).

Public Methods

Name	Description
 AddRelation	Adds a relation to the specified entities in the active sketch.
 DeleteAllRelations	Deletes all of the logical sketch relations in the sketch.
 DeleteRelation	Deletes the specified logical sketch relation in sketch.
 GetAllowedRelations	Gets the types of sketch relations valid for the specified entities.
 GetRelations	Gets all of the sketch relations in the sketch based on the specified filter.
 GetRelationsCount	Gets the number of sketch relations in the sketch based on the specified filter.
 IAddRelation	Adds a relation to the specified entities in the active sketch.
 IGetAllowedRelations	Gets the types of sketch relations valid for the specified entities.
 IGetAllowedRelationsCount	Gets the number of sketch relations valid for the specified entities.
 IGetRelations	Gets all of the sketch relations in the sketch based on the specified filter.

[Top](#)

 See Also

[ISketchRelationManager Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISketchRelation Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISketchRelationManager Interface Members

[See Also](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISketchRelationManager Interface

The following tables list the members exposed by [ISketchRelationManager](#).

 Public Methods

Name	Description
 AddRelation	Adds a relation to the specified entities in the active sketch.
 DeleteAllRelations	Deletes all of the logical sketch relations in the sketch.
 DeleteRelation	Deletes the specified logical sketch relation in sketch.
 GetAllowedRelations	Gets the types of sketch relations valid for the specified entities.
 GetRelations	Gets all of the sketch relations in the sketch based on the specified filter.
 GetRelationsCount	Gets the number of sketch relations in the sketch based on the specified filter.
 IAddRelation	Adds a relation to the specified entities in the active sketch.
 IGetAllowedRelations	Gets the types of sketch relations valid for the specified entities.
 IGetAllowedRelationsCount	Gets the number of sketch relations valid for the specified entities.
 IGetRelations	Gets all of the sketch relations in the sketch based on the specified filter.

[Top](#)

 See Also

[ISketchRelationManager Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISketchRelation Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddRelation Method (ISketchRelationManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRelationManager Interface](#) : AddRelation Method (ISketchRelationManager)

Adds a relation to the specified entities in the active sketch.

 .NET Syntax

Visual Basic (Declaration)

```
Function AddRelation( _  
    ByVal Entities As System.Object, _  
    ByVal RelationType As System.Integer _  
) As SketchRelation
```

Visual Basic (Usage)



```
Dim instance As ISketchRelationManager  
Dim Entities As System.Object  
Dim RelationType As System.Integer  
Dim value As SketchRelation  
  
value = instance.AddRelation(Entities, RelationType)
```

C#

```
SketchRelation AddRelation(  
    System.object Entities,  
    System.int RelationType  
)
```

C++/CLI

```
SketchRelation^ AddRelation(  
& System.Object^ Entities,  
& System.int RelationType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Entities

Array of entities to which add the relation

RelationType

Type of relation as defined by [swConstraintType_e](#)

Return Value

[Sketch relation](#)

Visual Basic for Applications (VBA) Syntax

See [SketchRelationManager::AddRelation](#).

Remarks

A sketch must be active for this method to have any effect. Use [IModelDoc2::GetActiveSketch2](#) or [IModelDoc2::GetActiveSketch2](#) to determine if a sketch is active.

See Also

[ISketchRelationManager Interface](#)
[ISketchRelationManager Members](#)
[ISketchRelationManager::AddRelation Method](#)
[ISketchRelationManager::GetRelationsCount Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

DeleteAllRelations Method (ISketchRelationManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRelationManager Interface](#) : DeleteAllRelations Method (ISketchRelationManager)

Deletes all of the logical sketch relations in the sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function DeleteAllRelations() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchRelationManager  
Dim value As System.Boolean  
  
value = instance.DeleteAllRelations()
```

C#

```
System.bool DeleteAllRelations()
```

C++/CLI

```
System.bool DeleteAllRelations();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if all of the logical sketch relations are deleted, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchRelationManager::DeleteAllRelations](#).

See Also

[ISketchRelationManager Interface](#)
[ISketchRelationManager Members](#)
[ISketchRelationManager::DeleteRelation Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

DeleteRelation Method (ISketchRelationManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRelationManager Interface](#) : DeleteRelation Method (ISketchRelationManager)

Deletes the specified logical sketch relation in sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function DeleteRelation(  
    ByVal ThisRelation As SketchRelation  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchRelationManager  
Dim ThisRelation As SketchRelation  
Dim value As System.Boolean  
  
value = instance.DeleteRelation(ThisRelation)
```

C#

```
System.bool DeleteRelation(  
    SketchRelation ThisRelation  
)
```

C++/CLI

```
System.bool DeleteRelation(  
& SketchRelation^ ThisRelation  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ThisRelation
[Sketch relation](#)

Return Value

True if the sketch relation is deleted, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchRelationManager::DeleteRelation](#).

See Also

[ISketchRelationManager Interface](#)
[ISketchRelationManager Members](#)
[ISketchRelationManager::DeleteAllRelations Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetAllowedRelations Method (ISketchRelationManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRelationManager Interface](#) : GetAllowedRelations Method (ISketchRelationManager)

Gets the types of sketch relations valid for the specified entities.

.NET Syntax

Visual Basic (Declaration)

```
Function GetAllowedRelations(  
    ByVal Entities As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchRelationManager  
Dim Entities As System.Object  
Dim value As System.Object  
  
value = instance.GetAllowedRelations(Entities)
```

C#

```
System.object GetAllowedRelations(  
    System.object Entities  
)
```

C++/CLI

```
System.Object^ GetAllowedRelations(  
& System.Object^ Entities  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Entities
Array of entities

Return Value

Array of sketch relation types valid for the specified entities as defined in [swConstraintType_e](#)

Visual Basic for Applications (VBA) Syntax

See [SketchRelationManager::GetAllowedRelations](#).

See Also

[ISketchRelationManager Interface](#)
[ISketchRelationManager Members](#)
[ISketchManagerRelation::GetAllowedRelations Method](#)
[ISketchManagerRelation::GetAllowedRelationsCount Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetRelations Method (ISketchRelationManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRelationManager Interface](#) : GetRelations Method (ISketchRelationManager)

Gets all of the sketch relations in the sketch based on the specified filter.

.NET Syntax

Visual Basic (Declaration)

```
Function GetRelations( _  
    ByVal Filter As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchRelationManager  
Dim Filter As System.Integer  
Dim value As System.Object
```

```
value = instance.GetRelations(Filter)
```

C#

```
System.object GetRelations(  
    System.int Filter  
)
```

C++/CLI

```
System.Object^ GetRelations(  
& System.int Filter  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Filter

Sketch relation as defined in [swSketchRelationFilterType_e](#)

Return Value

Array of [sketch relations](#)

Visual Basic for Applications (VBA) Syntax

See [SketchRelationManager::GetRelations](#).

Example

[Replace Sketch Relation \(VBA\)](#)
[Get Sketch Relations \(VBA\)](#)
[Get Sketch Relations \(VB.NET\)](#)
[Get Sketch Relations \(C#\)](#)

Remarks

Because some of the objects may be NULL, you should check for this before accessing them. You should deallocate on any dynamically allocated memory.

See Also

[ISketchRelationManager Interface](#)
[ISketchRelationManager Members](#)
[ISketchRelationManager::IGetRelations Method](#)
[ISketchRelationManager::GetRelationsCount Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetRelationsCount Method (ISketchRelationManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRelationManager Interface](#) : GetRelationsCount Method (ISketchRelationManager)

Gets the number of sketch relations in the sketch based on the specified filter.

.NET Syntax

Visual Basic (Declaration)

```
Function GetRelationsCount( _  
    ByVal Filter As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketchRelationManager  
Dim Filter As System.Integer  
Dim value As System.Integer  
  
value = instance.GetRelationsCount(Filter)
```

C#

```
System.int GetRelationsCount(  
    System.int Filter  
)
```

C++/CLI

```
System.int GetRelationsCount(  
& System.int Filter  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Filter
Sketch relation as defined in [swSketchRelationFilterType_e](#)

Return Value

Number of sketch relations

Visual Basic for Applications (VBA) Syntax

See [SketchRelationManager::GetRelationsCount](#).

Example

[Replace Sketch Relation \(VBA\)](#)

Remarks

Call this method before calling [ISketchRelationManager::IAddRelation](#) and [ISketchRelationManager::IGetRelations](#) to get the size of the array for that method.

See Also

[ISketchRelationManager Interface](#)
[ISketchRelationManager Members](#)
[ISketchRelationManager.AddRelation Method](#)
[ISketchRelationManager.GetRelations Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IAddRelation Method (ISketchRelationManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRelationManager Interface](#) : IAddRelation Method (ISketchRelationManager)

Adds a relation to the specified entities in the active sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function IAddRelation( _  
    ByVal NumEntities As System.Integer, _  
    ByRef EntityArray As System.Object, _  
    ByVal RelationType As System.Integer _  
) As SketchRelation
```

Visual Basic (Usage)



```
Dim instance As ISketchRelationManager  
Dim NumEntities As System.Integer  
Dim EntityArray As System.Object  
Dim RelationType As System.Integer  
Dim value As SketchRelation  
  
value = instance.IAddRelation(NumEntities, EntityArray, RelationType)
```

C#

```
SketchRelation IAddRelation(  
    System.int NumEntities,  
    ref System.object EntityArray,  
    System.int RelationType  
)
```

C++/CLI

```
SketchRelation^ IAddRelation(  
& System.int NumEntities,  
& System.Object^% EntityArray,  
& System.int RelationType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumEntities

Number of entities to which to add the relation
EntityArray
Array of entities to which to add the relation
RelationType
Type of relation as defined by [swConstraintType_e](#)

Return Value

[Sketch relation](#)

Visual Basic for Applications (VBA) Syntax

See [SketchRelationManager::IAddRelation](#).

Remarks

A sketch must be active for this method to have any effect. Use [IModelDoc2::GetActiveSketch2](#) or [IModelDoc2::IGetActiveSketch2](#) to determine a sketch is active.

Call [ISketchRelationManager::GetRelationsCount](#) before calling this method to get the value for NumEntities.

See Also

[ISketchRelationManager Interface](#)
[ISketchRelationManager Members](#)
[ISketchRelationManager::AddRelation Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetAllowedRelations Method (ISketchRelationManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRelationManager Interface](#) : IGetAllowedRelations Method (ISketchRelationManager)

Gets the types of sketch relations valid for the specified entities.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetAllowedRelations( _  
    ByVal NumEntities As System.Integer, _  
    ByRef EntityArray As System.Object, _  
    ByVal NumAllowedRelations As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketchRelationManager  
Dim NumEntities As System.Integer  
Dim EntityArray As System.Object  
Dim NumAllowedRelations As System.Integer  
Dim value As System.Integer  
  
value = instance.IGetAllowedRelations(NumEntities, EntityArray, NumAllowedRelations)
```

C#

```
System.int IGetAllowedRelations(  
    System.int NumEntities,  
    ref System.object EntityArray,  
    System.int NumAllowedRelations  
)
```

C++/CLI

```
System.int IGetAllowedRelations(
& System.int NumEntities,
& System.Object^% EntityArray,
& System.int NumAllowedRelations
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumEntities

Number of entities

EntityArray

Array of entities

NumAllowedRelations

Number of relations valid for the specified entities

Return Value

- in-process, unmanaged C++: Pointer to an array of sketch relation types valid for the specified entities as defined in [swConstraintType_e](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Call [ISketchRelationManager::IGetAllowedRelationsCount](#) before calling this method to get the value for NumEntities.

See Also

[ISketchRelationManager Interface](#)

[ISketchRelationManager Members](#)

[ISketchRelationManager::GetAllowedRelations Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetAllowedRelationsCount Method (ISketchRelationManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRelationManager Interface](#) : IGetAllowedRelationsCount Method (ISketchRelationManager)

Gets the number of sketch relations valid for the specified entities.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetAllowedRelationsCount( _
    ByVal NumEntities As System.Integer, _
    ByRef EntityArray As System.Object _
) As System.Integer
```

Visual Basic (Usage)

```
Dim instance As ISketchRelationManager
Dim NumEntities As System.Integer
Dim EntityArray As System.Object
Dim value As System.Integer
```

```
value = instance.IGetAllowedRelationsCount(NumEntities, EntityArray)
```

C#

```
System.int IGetAllowedRelationsCount(
    System.int NumEntities,
    ref System.object EntityArray
)
```

C++/CLI



```
System.int IGetAllowedRelationsCount(
& System.int NumEntities,
& System.Object^% EntityArray
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumEntities
Number of entities
EntityArray
Array of entities

Return Value

Number of sketch relations valid for the specified entities

Visual Basic for Applications (VBA) Syntax

See [SketchRelationManager::IGetAllowedRelationsCount](#).

Remarks

Call this method before calling [ISketchRelationManager::IGetAllowedRelations](#) to get the size of the array for that method.

See Also

[ISketchRelationManager Interface](#)
[ISketchRelationManager Members](#)
[ISketchRelationManager::GetAllowedRelations Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetRelations Method (ISketchRelationManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRelationManager Interface](#) : IGetRelations Method (ISketchRelationManager)

Gets all of the sketch relations in the sketch based on the specified filter.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetRelations( _
    ByVal Filter As System.Integer, _
    ByVal Count As System.Integer _
) As SketchRelation
```

Visual Basic (Usage)



```
Dim instance As ISketchRelationManager
Dim Filter As System.Integer
Dim Count As System.Integer
Dim value As SketchRelation
```

```
value = instance.IGetRelations(Filter, Count)
```

C#

```
SketchRelation IGetRelations(
    System.int Filter,
    System.int Count
)
```

C++/CLI

```
SketchRelation^ IGetRelations(  
& System.int Filter,  
& System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Filter

Sketch relation as defined in [swSketchRelationFilterType_e](#)

Count

Number of sketch relations

Return Value

- in-process, unmanaged C++: Pointer to an array of [sketch relations](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Because some of the objects may be NULL, you should check for this before accessing them. You should deallocate on any dynamically allocated memory.

Call [ISketchRelationManager::GetRelationsCount](#) before calling this method to get the value for Count.

See Also

[ISketchRelationManager Interface](#)

[ISketchRelationManager Members](#)

[ISketchRelationManager::GetRelations Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0